

1. Yes, $x_1^* = 0, x_2^* = 3$ is a feasible solution. The verification is done by substitution:

$$0 + 3 = 3 \quad \checkmark$$

$$0 \geq 0 \quad \checkmark$$

$$3 \geq 0. \quad \checkmark$$

The first two constraints are active at $\{x_1^*, x_2^*\}$. In particular, notice that equality constraints are always active when verified. The point is labeled A in the figure below.

2. No, $x_1^* = 2, x_2^* = 2$ is not a feasible solution, since the equality constraint is not verified:

$$2 + 2 = 3 \quad \times$$

$$2 \geq 0 \quad \checkmark$$

$$2 \geq 0. \quad \checkmark$$

The point is labeled B in the figure below.

The feasible region for the problem is the segment represented in bold in the following graph.

